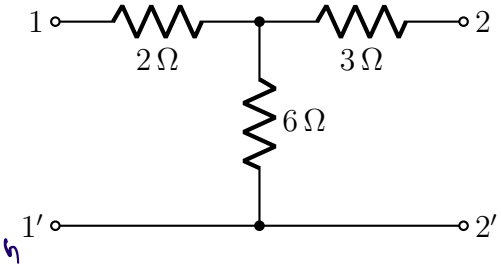
EE2015: Electric Circuits and Networks

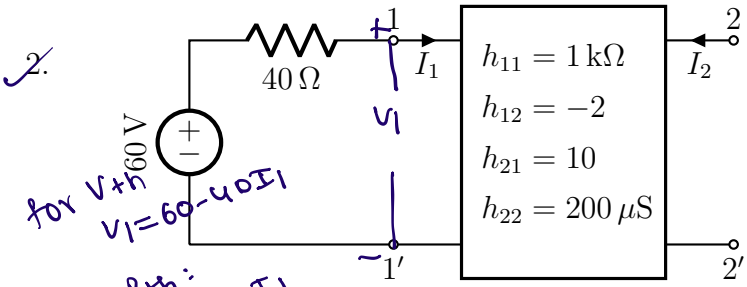
Tutorial 5

(September 19, 2025)

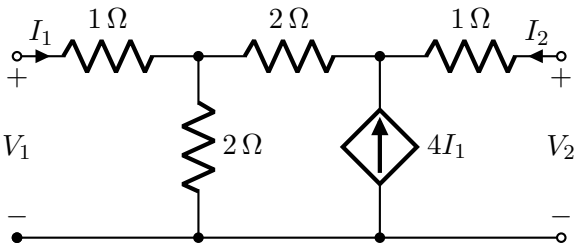
1. Find the h-parameters of the circuit shown on the right.



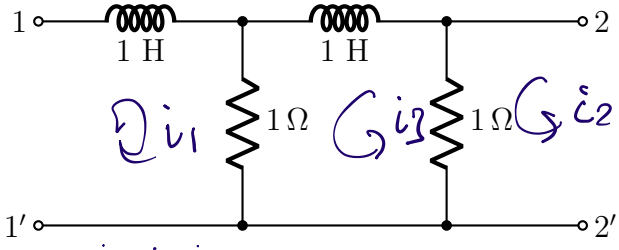
2. The h-parameters of a two-port network are as shown on the left. Find the Thevenin equivalent as seen at the output terminal pair 22′.



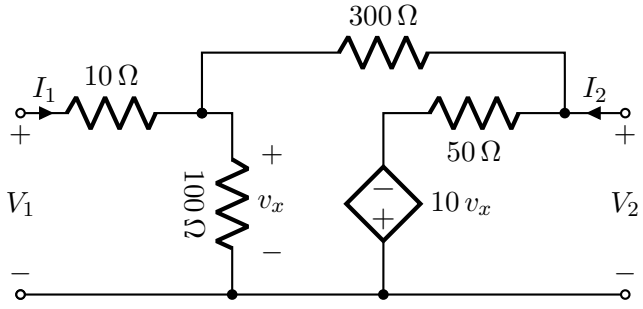
3. Find the h-parameters of the two-port network shown on the right.



4. Find the ladder network shown on the left, determine the g-parameters in the s-domain.



5. Determine the h- and g-parameters of the two-port network shown on the right

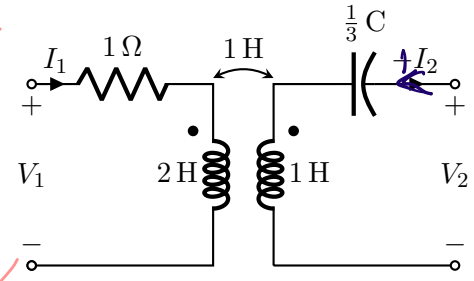


$V_1 = I_1 + 2SI_1 + SI_2$

$V_2 = \frac{3}{S} I_2 + SI_1 + SI_2$

$I_1 = \frac{V_2}{S} + -I_2\left(1 + 3/S^2\right)$

6.



Find the *abcd* parameters of the two-port network shown on the left.

7. The T-parameters of a network are as shown below. The output port is connected to a variable resistor R_L and adjusted for maximum power transfer. Find R_L and the value of the maximum power transferred.

$V_1 = 4V_2 - 20I_2$

$I_1 = V_2/10 - 2I_2$

$V_1 = 50 - 10I_1$

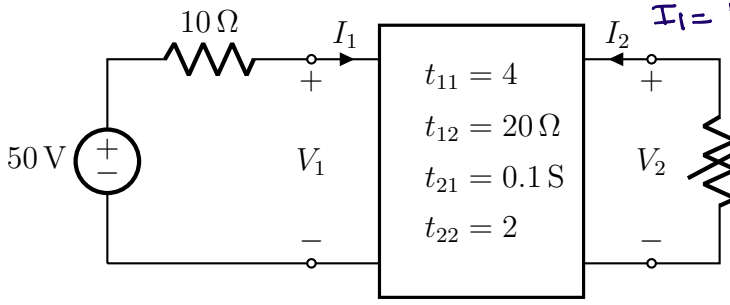
$V_2 = -I_2 R$

$I_1 = -I_2\left[2 + R/10\right]$

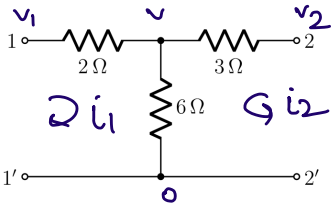
$50 + I_2[20 + R] = 4I_2 R - 20I_2$

$\Rightarrow I_2[-4R - 20 - 20 - R] = 50$

$I_2 = -\frac{50}{5R + 40} = -\left(\frac{10}{R+8}\right)$



1. Find the h-parameters of the circuit shown on the right.



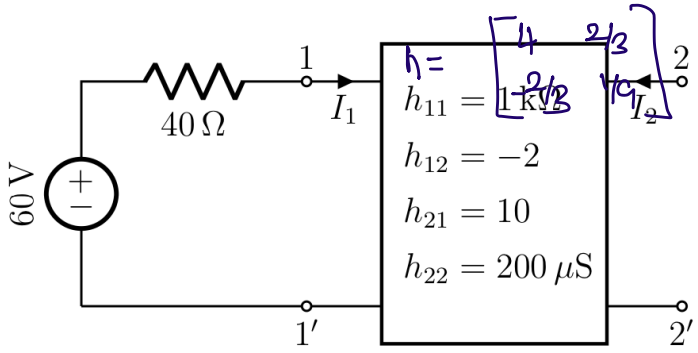
$v_1 - v = 2i_1 \quad v_2 - v = 3i_2 \Rightarrow v_1 = 8i_1 + 6i_2 = 8i_1 + 6(v_2 - 6i_1) = 4i_1 + 2v_2$

$v = 6(i_1 + i_2)$

$v_2 = 6i_1 + 9i_2$

$i_2 = \frac{v_2}{9} - \frac{6i_1}{9}$

2.



The h-parameters of a two-port network are as shown on the left. Find the Thevenin equivalent as seen at the output terminal pair 22′.

$h = \begin{bmatrix} 4 & 2/3 \\ -2/3 & 1/9 \end{bmatrix}$

$V_1 = h_{11} I_1 + h_{12} V_2 \Rightarrow V_1 = 1000 I_1 - 2V_2$

$I_1 = h_{21} I_1 + h_{22} V_2 \Rightarrow I_2 = 10 I_1 + V_2/500$

$\Rightarrow 0 = 10I_1 + V_2/500 \Rightarrow I_1 = -\frac{V_2}{5000}$

$60 - 40I_1 = 1000I_1 - 2V_2 = -\frac{11V_2}{5} \Rightarrow 60 = \frac{V_2}{125} - \frac{11V_2}{5} = -\frac{274}{125}V_2$

$\Rightarrow V_2 = -27.37\text{V}$

$R_{th} \rightarrow 60\text{V}$ SC

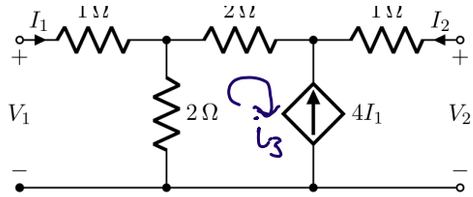
$-40I_1 = 1000I_1 - 2V_2 \Rightarrow V_2 = 502I_1$

$R_{th} = \frac{502}{11.04} = 45.47\,\Omega$

$I_2 = 10I_1 + 1.04I_1 = 11.04I_1$

3.

Find the *h*-parameters of the two-port network shown on the right.



$V_1 = i_1 + 2(i_1 - i_3) = 3i_1 - 2i_3 \qquad i_3 + i_2 = -4i_1$

$V_2 = i_2 - 2i_3 + 2(i_1 - i_3) = 2i_1 + i_2 - 4i_3$

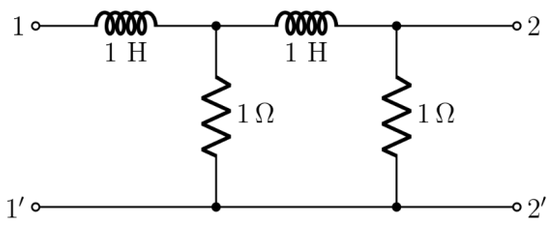
$V_1 = 3i_1 - 2(-4i_1 - i_2) = 11i_1 + 2i_2$

$V_2 = 2i_1 + i_2 - 4(-4i_1 - i_2) = 18i_1 + 5i_2 \qquad i_2 = \frac{V_2 - 18i_1}{5}$

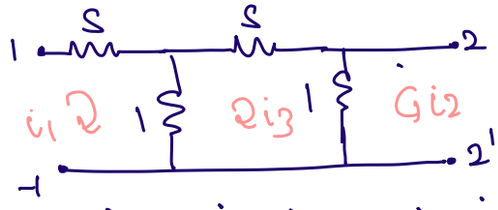
$\Rightarrow V_1 = 11i_1 + \frac{2V_2}{5} - \frac{36i_1}{5} = \frac{19i_1}{5} + \frac{2V_2}{5}$

$i_2 = -\frac{18}{5}i_1 + \frac{V_2}{5}$

4.



Find the ladder network shown on the left, determine the *g*-parameters in the *s*-domain.



$V_1 - Si_1 - 1(i_1 - i_3) = 0$

$V_1 = (S+1)i_1 - i_3$

$V_2 = i_2 + i_3$

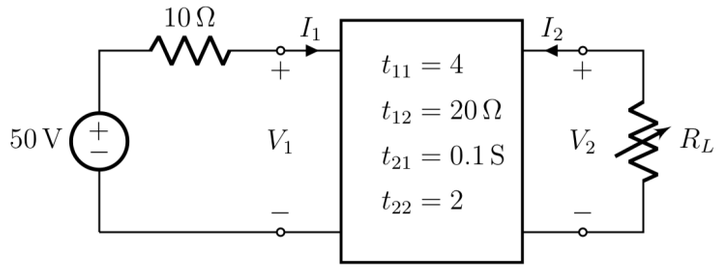
$Si_3 + (i_3 + i_2) + (i_3 - i_1) = 0 \Rightarrow (S+2)i_3 = i_1 - i_2$

$$V_1 = (S+1)i_1 - \frac{i_1 - i_2}{S+2} = \frac{(S^2+3S+1)i_1 + i_2}{S+2} \Rightarrow \frac{(S+2)V_1 - i_2}{S^2+3S+1} = i_1$$

$$V_2 = i_2 + \frac{i_1 - i_2}{S+2} = \frac{(S+1)i_2 + i_1}{S+2} = \left(\frac{S+1}{S+2}\right)i_2 + \frac{V_1}{S^2+3S+1} - \frac{i_2}{(S^2+3S+1)(S+2)}$$

$$= \frac{V_1}{S^2+3S+1} + \frac{S^3+4S^2+4S}{(S^2+3S+1)(S+2)} i_2$$

7. The T-parameters of a network are as shown below. The output port is connected to a variable resistor R_L and adjusted for maximum power transfer. Find R_L and the value of the maximum power transferred.



$V_1 = 4V_2 - 20I_2$ $\qquad V_1 = 50 - 10I_1$

$I_1 = 0.1V_2 - 2I_2$ $\qquad V_2 = -I_2R_L$

$50 - 10I_1 = 4V_2 - 20I_2 \Rightarrow 25 - 5I_1 = 2V_2 - 10I_2$

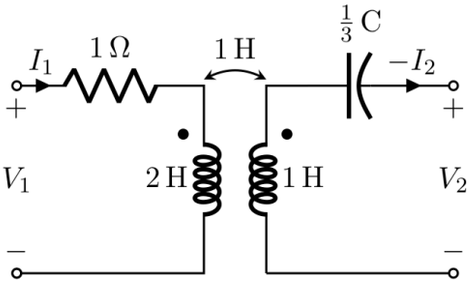
$V_2 = -I_2R_L \Rightarrow 25 - 5I_1 = -2I_2R_L - 10I_2$

$I_1 = 0.1\{-I_2R_L\} - 2I_2 = I_2\{-2 - 0.1R_L\}$

$\Rightarrow 25 + 10I_2 + 0.5I_2R_L = -2I_2R_L - 10I_2$

$\Rightarrow 25 = I_2[-2.5R_L - 20] \Rightarrow I_2 = \dfrac{-25}{2.5R_L + 20} = \dfrac{-10}{R_L + 8}$

$P = 100\left(\dfrac{R_L}{(R_L+8)^2}\right) \Rightarrow R_L = 8\ ;\ P = 100/32 = 3.125$

6. 

Find the *abcd* parameters of the two-port network shown on the left.

$V_1 = I_1 + 2sI_1 + sI_2$

$V_2 = \dfrac{3}{s}I_2 + sI_2 + sI_1$

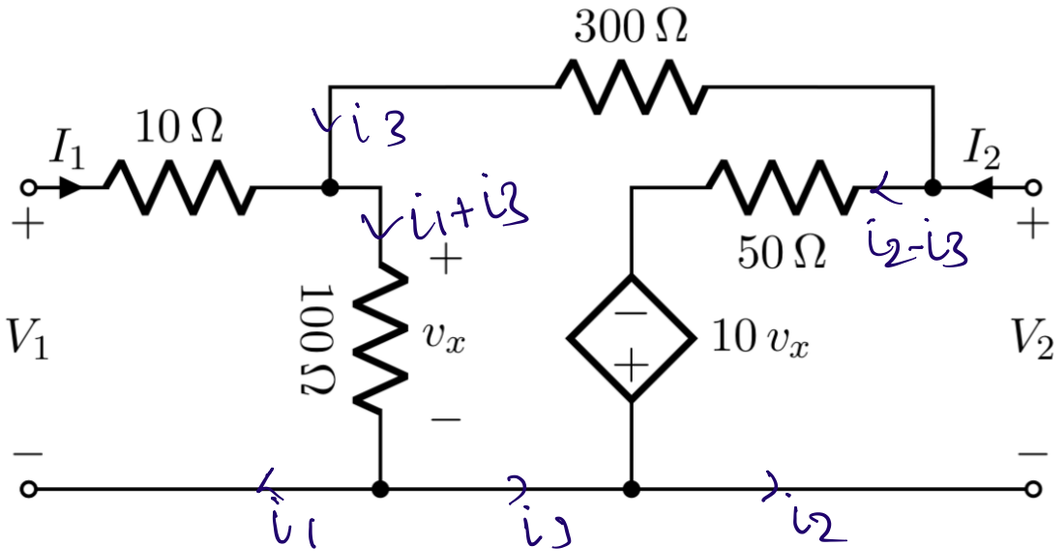
$I_1 = \dfrac{V_2}{s} - \left(\dfrac{3}{s^2} + 1\right)I_2$

$V_1 = \left[\dfrac{V_2}{s} - \left(\dfrac{3}{s^2} + 1\right)I_2\right][1 + 2s] + sI_2$

$= V_2\dfrac{[1+2s]}{s} + I_2\left[s - \dfrac{3+s^2}{s^2}(1+2s)\right]$

$= V_2\left[2 + \dfrac{1}{s}\right] + I_2\left[-\dfrac{3}{s^2} - \dfrac{6}{s} - 1 - s\right]$

$\begin{bmatrix} 2 + 1/s & 3/s^2 + 6/s + 1 + s \\ 1/s & 1 + 3/s^2 \end{bmatrix}$



$V_1 = 10I_1 + 100(i_1+i_3) = 0$ $\qquad v_x = 100(i_1+i_3)$

$V_2 = 50(i_2-i_3) - 10v_x = 50(i_2-i_3) - 1000(i_1+i_3)$

$300\,i_3 + 1100(i_1+i_3) + 50(i_3-i_2) = 0$

$i_3[1450] = 50i_2 - 1100i_1$

$$\boxed{i_3 = \frac{i_2}{29} - \frac{22}{29}i_1}$$

$V_1 = 110\,i_1 + 100\left(\frac{i_2}{29} - \frac{22}{29}i_1\right)$ $\qquad V_2 = 50i_2 - 1000i_1 - 1050\left(\frac{i_2}{29} - \frac{22}{29}i_1\right)$

$V_1 = \frac{990}{29}i_1 + \frac{100}{29}i_2$ $\qquad V_2 = -\frac{5900}{29}i_1 + \frac{400}{29}i_2$

$$i_2 = \frac{29}{400}v_2 + \frac{59}{4}i_1$$